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AGE 418
Sensors and Controls in Precision Agriculture

Lecture 10 Compendium:
Data Management and Analysis
Turning Raw Data into Decisions

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1 1. The Data Lifecycle

Precision Agriculture generates massive amounts of data. A single yield monitor can log 5,000 data points per hectare. Managing this "Big Data" is an engineering challenge.

1.1 The 4 Stages

1. **Collection:** Sensors capture raw signals (Voltage, Resistance, GPS coordinates).
2. **Cleaning:** Removing errors, outliers, and noise. "Garbage In, Garbage Out."
3. **Analysis:** Converting points into maps (Interpolation) and finding relationships (Correlation).
4. **Archiving:** Storing data securely for future use (Historical trends).

2 2. File Formats and Compatibility

2.1 1. Proprietary Formats

Every manufacturer used to have a secret language.

- John Deere (.GSW, .JDL)
- Case IH (.CN1)
- Trimble (.AGDATA)

Problem: You cannot open a John Deere file in Case IH software.

2.2 2. The Shapefile (.shp)

The universal standard for spatial vector data.

- Developed by ESRI but now open.
- It is actually a **collection of files** (must be kept together):
  - **.shp:** The geometry (shapes).
  - **.shx:** The index (search speed).
  - **.dbf:** The database (attributes like "Yield = 5 tons").
  - **.prj:** The projection (Coordinate system).

2.3 3. ISOXML (ISO 11783)

The modern agricultural standard designed to replace Shapefiles. It handles complex tasks like Variable Rate prescriptions better than shapefiles.

3 3. Data Cleaning

Before analyzing data, it must be cleaned. Up to 30% of raw yield data is erroneous.

3.1 Common Filters

1. **Flow Delay (Lag):** Correcting the time shift between header and sensor.
2. **Moisture Limits:** Deleting points where moisture is 0% or >40% (sensor error).
3. **Yield Limits:** Deleting points where yield is > 20 tons/ha (biological impossibility for maize).
4. **Start/Stop Delays:** Removing data points recorded while the combine was ramping up speed or slowing down.
5. **GPS Wander:** Removing points that drift outside the field boundary.

4 4. Spatial Interpolation

Sensors collect **Points** (discrete data). To make a map, we need a **Surface** (continuous data). The process of estimating values between known points is called **Interpolation**.

4.1 1. Inverse Distance Weighting (IDW)

- **Logic:** "Things that are close to one another are more alike than things that are far apart."
- **Math:** The value of an unknown point is the weighted average of nearby known points. The weight drops rapidly as distance increases.
- **Use:** Good for dense data like Yield Maps (thousands of points).

4.2 2. Kriging (Geostatistics)

- **Logic:** Uses statistical models (Semivariogram) to predict values based on spatial auto-correlation.
- **Math:** More complex than IDW. Provides an "Error Map" showing how confident the prediction is.
- **Use:** Best for sparse data like Soil Sampling (only 1 point per hectare).

Engineering Choice

If you have **Dense Data** (Yield Monitor), use **IDW** (Faster). If you have **Sparse Data** (Soil Lab Results), use **Kriging** (More accurate).

5 5. Farm Management Information Systems (FMIS)

An FMIS is the software hub where data is stored, visualized, and processed.

5.1 Types of FMIS

- **Desktop Software:** Installed on a PC (e.g., SMS Advanced, QGIS). Powerful, works offline, but data is stuck on one computer.
- **Cloud Platforms:** Web-based (e.g., John Deere Ops Center, Climate FieldView). Accessible from phone/tablet anywhere. Requires internet.

5.2 Functions of FMIS

1. **Record Keeping:** What did I plant? When did I spray? 2. **Mapping:** Visualizing yield and soil maps. 3. **Prescription Writing:** Creating Rx maps for VRT. 4. **Profitability Analysis:** Overlaying Yield Map (Income) vs Cost Map (Expense) to see Profit/Loss per acre.

6 6. Data Ownership and Privacy

As agriculture becomes digital, legal and ethical questions arise.

6.1 Who owns the data?

- **The Farmer?** (It's their land and crop).
- **The Machine Manufacturer?** (Their sensor collected it).
- **The Cloud Provider?** (Their server stores it).

Current Standard: The Farmer owns the raw data, but often signs a license allowing the company to use "anonymized aggregated data" for research.

6.2 Security Risks

- **Market Manipulation:** If a company knows real-time yields of 50% of farms, they can manipulate grain prices.
- **Cybersecurity:** Hackers locking tractors (Ransomware) or stealing prescription maps.

7 7. Lecture Assessment

7.1 Part A: Standard Concept Questions

1. Define the **Data Lifecycle** in Precision Agriculture.
2. Why is **Data Cleaning** necessary before generating a yield map? Give two examples of "dirty data."
3. Explain the structure of a **Shapefile**. Why must you always keep the .shp, .shx, and .dbf files together?
4. What is the difference between **Proprietary** formats and **Standardized** formats (ISOXML)? Why is standardization important for farmers?
5. Define **Spatial Interpolation**. Why do we need it?
6. Compare **IDW** and **Kriging**. Which method would you choose for soil sample data, and why?
7. What is an **FMIS**? List three primary functions it performs.
8. Discuss the pros and cons of **Cloud-Based FMIS** vs **Desktop FMIS** in the context of rural Tanzania.
9. Explain the concept of **Data Ownership** in agriculture. Why are farmers concerned about sharing their data?
10. How can an FMIS be used to create a **Profit Map**? What two layers are required?

7.2 Part B: Advanced Analysis (Open Source Data)

These questions emphasize the use of open-source datasets available for Africa.

1. **QGIS Interpolation Exercise:** Download a sample CSV file containing "Latitude, Longitude, Soil_pH" (you can create a dummy one with 20 points). *Task:* Load this into **QGIS** (Open Source GIS). Perform an **IDW Interpolation** to create a continuous pH map. Explain how changing the "Distance Coefficient" (Power) affects the smoothness of the map.
2. **Standardizing Data (GDAL):** You have a raster image in a proprietary format (e.g., .img) and need it in GeoTIFF for an open-source viewer. *Task:* Investigate the **GDAL** (Geospatial Data Abstraction Library) command line tools. Write the specific command line instruction to convert `input.img` to `output.tif`.
3. **Cloud Data Privacy Policy Analysis:** Search for the "Data Privacy Policy" of a major digital ag platform operating in Africa (e.g., **FarmView**, **DigiFarm**, or **John Deere**). *Task:* Read the section on data ownership. Does the farmer retain ownership? Can the company sell the aggregated data? Summarize the terms in simple language.
4. **Building a Simple FMIS Database:** Using **Microsoft Excel** or **Google Sheets**. *Task:* Design a table structure for a "Field Record Log." What columns (attributes) are mandatory to make the data spatially relevant? (Hint: Field ID, Crop, Date, Input Type, Rate, Operator). Why is a standardized dropdown menu better than free text entry?

5. **OpenStreetMap (OSM) Extraction:** Go to **export.hotosm.org** (Humanitarian OpenStreetMap Team). *Task:* Select a region in **Dodoma**. Download the data as a **Shapefile**. Open the folder. Identify the '.shp', '.shx', and '.dbf' files. Open the '.dbf' in Excel to see the attributes. How could this vector data be used to plan logistics for a tractor fleet?